Table 5.2: Basic Approaches to Food preservation

S.No	Approach	Examples of process
1.	Removal of microorganisms	Avoidance of microbial contamination, physical filtration, centrifugation.
2.	Low temperature	Refrigeration, Freezing
3.	High temperature	Partial or complete heat inactivation of microorganisms (pasteurization and canning
4.	Reduced water availability	Water removal, as with Lyophilization or freeze drying use of spray dryers or heating drums decreasing water availability by addition of solutes such as salt or sugar.
5.	Chemical – based preservation	Addition of specific inhibitory compounds (Example: organic acids, nitrates, sulfur dioxide)
6.	Radiation	Use of ionizing (gamma rays) and non ionizing (UV) radiation
7.	Microbial product – based inhibition	The addition of substances such as bacteriocins to foods to control food – borne pathogens

Disadvantages

- Excess salt and sugar are used in the preservation of food which is not good for health.
- Some methods of food preservation may lead to loss of nutrients.

Methods of food preservation

Principles of Food preservation

In accomplishing the preservation of foods by the various methods, the following principles are involved.

1. Prevention or delay of microbial decomposition.
   a. By keeping out microorganism (asepsis)
   b. By removal of microorganism. Example: Filtration
   c. By hindering the growth and activity of microorganism Example: Low temperature, drying, anaerobic conditions or chemicals.
   d. By killing the microorganism Example: Heat or radiation

Infobits

"Typhoid Fever and Canned Meat"

Minor errors in canning have led to major typhoid outbreaks. In 1964 canned beef produced in South America was cooled, after sterilization with non chlorinated water. The vacuum created when the cans were cooled drew *Salmonella typhi* into some of the cans, which were not completely sealed. This contaminated product was later sliced in an Aberdeen, Scotland, Food store and the meat slicer became a continuing contamination source the result was a major epidemic that involved 400 people. The *Salmonella typhi* was a South American strain and eventually the contamination was traced to the contaminated water used to cool the cans. This emphasizes the importance of careful food processing and handling to control the spread of disease during food production and preparation.



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